SE(3)-WAM: Execution-Time-Aligned World-Action Modeling for Dynamic Manipulation

Anonymous Authors

***Abstract*— Dynamic manipulation is not solved merely by reacting faster: camera exposure, state estimation, policy inference, communication, and actuation make the scene observed by a robot stale when an action takes effect. We present SE(3)-WAM, an execution-time-aligned world-action model for manipulating exogenously moving rigid objects. From timestamped two-view RGB-D, proprioceptive, and action history, SE(3)-WAM predicts target-centric multi-horizon $\mathrm{SE}(3)$, dense metric 3-D scene flow, and uncertainty at the actual action-execution times. A fixed-size physical summary is consumed through a one-way cross-attention bridge by a flow-matching action expert, while remaining independently measurable and directly intervenable. We also introduce DB0, a replayable dynamic-manipulation protocol with exact-current counterfactual histories, latency/TTC sweeps, an orientation causal split, physical occlusion, and contact-driven tasks. On the DB0 Core, SE(3)-WAM obtains [XX.X]% success versus [XX.X]% for a matched history-aware policy ([+X.X] percentage points; 95% CI [X.X, X.X]). Forecast swapping changes success by [-X.X] points while holding the current observation fixed, testing whether action generation causally uses the predicted future. Native external and real-conveyor results are [pending frozen external and real-robot evaluation]. These experiments separate better temporal representations from actionable, physically correct foresight.**

I. Introduction

Robots manipulating moving objects act on a delayed view of the world. An RGB-D frame is exposed and transferred, a policy runs, a command is queued, and the controller finally applies it. The target can translate, rotate, disappear behind an occluder, or undergo a contact event during this interval. Faster inference and continuous action streaming reduce the delay [1], [2], but cannot make a nonzero physical pipeline instantaneous or recover motion hidden during occlusion. The relevant question is therefore not only *where is the object now?*, but *what physical state will be present when this command is applied?*

Existing work attacks complementary pieces of this problem. DynamicVLA streams actions from sparse history but has no explicit environmental future [1]. PUMA predicts future features only as training supervision [3]; DBC-TFP and AHEAD explicitly feed predicted images or latent tokens to a policy [4], [5]. PhysMani predicts 3-D Gaussian motion and makes it available to an action policy, but requires per-scene reconstruction and online optimization [6]. LaMP instead transfers a learned 3-D scene-flow prior to an action expert [7]. These advances mean that "future-aware action" is no longer, by itself, a defensible novelty claim. A remaining intersection is a feed-forward system that (i) infers exogenous motion from timestamped raw RGB-D history, (ii) exposes target-centric metric 3-D state at measured execution times, (iii) quantifies uncertainty, and (iv) makes the world-to-action path causally auditable under matched compute and data.

We introduce **SE(3)-WAM**, an action-centered world-action model built around this intersection (Fig. 1). A temporal 3-D encoder lifts frozen visual features with measured depth and camera geometry. A world expert predicts future target transforms on $\mathrm{SE}(3)$, dense scene displacement, and heteroscedastic uncertainty at eight physical-time horizons. Rather than sending all spatial tokens to the policy, a mover- and target-aware reducer forms a small set of physical summary tokens. A one-way bridge conditions an action expert on these tokens; zeroing, swapping, noising, or replacing them with oracle summaries directly tests whether the action depends on the predicted future. Continuous execution discards stale action prefixes and replaces overlapping commands with the newest temporally valid chunk.

The paper makes three contributions:

- an execution-time-aligned world-action formulation that predicts target-centric multi-horizon $\mathrm{SE}(3)$, dense metric 3-D scene flow, and uncertainty from non-privileged timestamped RGB-D history;
- a fixed-size, one-way physical interface from forecast to action, paired with exact-current and forecast-intervention tests that distinguish auxiliary representation learning from inference-time use of foresight; and
- DB0, a physical-clock evaluation protocol spanning interception, moving-goal release, rotational phase, gravity/friction motion, contact-triggered replanning, external benchmarks, static compatibility, and a real Franka conveyor.

The design deliberately does not claim that a generic world model or 3-D future is unprecedented. Its empirical claim is narrower and falsifiable: under exogenous motion and real execution delay, a correct metric future interface must improve behavior over parameter-, data-, and latency-matched history-aware policies, and corrupting that interface must degrade action in the predicted direction.

II. Related Work

A. Dynamic Manipulation and Real-Time Execution

Classical dynamic grasping predicts target pose and combines it with reachability-aware planning [8]; visual servo and tracking systems can instead make a moving target locally quasi-static. GEM applies this modular strategy to conveyor manipulation at scale [9]. Recent learned policies reduce observation staleness through action chunking and asynchronous execution. DynamicVLA combines two-frame history, continuous inference, and latent-aware action

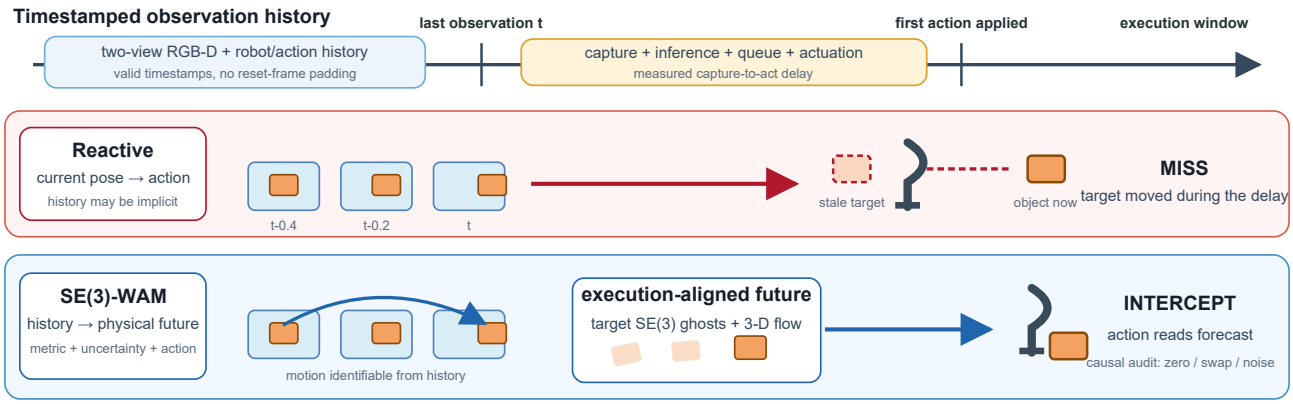


Fig. 1. Execution-time-aligned dynamic manipulation. A reactive policy targets the last observed pose and misses after the capture-to-act delay ℓ_t. SE(3)-WAM uses timestamped history to predict metric target $\mathrm{SE}(3)$, dense 3-D motion, and uncertainty at the actual execution window, then exposes that prediction through a compact, intervenable interface to the action policy.

streaming [1]; VLASH rolls robot state forward to align asynchronous chunks [2], while AHA-WAM separates a slower world planner from a faster action process [10]. These are strong alternatives, not straw-man baselines: they test whether explicit environmental foresight adds value beyond a well-engineered reactive loop.

B. Future Supervision and Forecast-to-Action Models

World-action models often use future prediction to shape representations without executing the predicted future. PUMA uses historical optical flow and world queries to predict pooled future DINO features during training [3]; GaussianDream and EgoWAM study 3-D or geometric future supervision that can be removed at inference [11], [12]. Fast-WAM further shows that co-training can outperform test-time imagination in some regimes [13]. Such results motivate our matched *joint auxiliary* control.

Other methods make foresight an inference-time input. DBC-TFP fuses a predicted future RGB image with the current image [4]. AHEAD predicts selected future OpenVLA patch tokens using 2-D flow-derived kinematics and uncertainty-aware horizon stopping [5]. LaWAM conditions action on latent future subgoals [14], while FutureRTC anticipates the execution-time latent state [15]. Making Foresight Actionable demonstrates why visual plausibility alone is insufficient and motivates causal intervention on the world-action interface [16]. OA-WAM's addressable slots and binding interventions make a complementary argument for explicit world–action interfaces [17]. Our interface is designed for a metric, execution-time version of this audit.

Action chunking and conditional generative policies provide the control substrate rather than the paper's novelty. ACT, Diffusion Policy, and DP3 establish strong sequence and metric-3-D action baselines [18]–[20]. We therefore match the action objective, chunk, and 3-D input before attributing gains to the predicted future.

C. Metric 3-D World-Action Modeling

PhysMani forecasts a divergence-free velocity field over canonical 3-D Gaussians and conditions a policy through local cross-attention [6]. LaMP uses 3-D scene-flow hidden states as a latent motion prior [7]; PointWorld predicts action-conditioned dense 3-D point flow for model-predictive control [21]; object-centric Gaussian models forecast multiple rigid-body transforms under candidate actions [22]. 3DFlowAction predicts desired object flow as an action representation rather than exogenous evolution [23]. SE(3)-WAM differs in the combination of raw timestamped RGB-D history, target-centric metric $\mathrm{SE}(3)$, dense flow, measured execution horizons, uncertainty, and a compact intervenable bridge. Table I summarizes this scope rather than asserting priority for any single ingredient.

III. SE(3)-WAM

A. Physical-Time Problem Formulation

At policy query t, the robot receives a masked, timestamped history $\mathcal{H}_t$ of two-view RGB-D observations, camera calibration, proprioception, and issued/applied actions. The most recent sensor timestamp is τ_t^s. Telemetry estimates the capture-to-act delay ℓ_t, including inference, serialization, queuing, and actuation. The h-th world target is evaluated at

$$\tau_h^\star = \tau_t^s + \ell_t + h\Delta_a, \qquad h = 0, \dots, H-1, \tag{1}$$

where Δ_a is the controller period. Thus a fixed model index denotes a physical execution time, not merely "h video frames ahead." The model predicts the task-object transform $\hat{\boldsymbol{T}}_h \in \mathrm{SE}(3)$, patch displacement $\hat{\boldsymbol{f}}_{i,h} \in \mathbb{R}^3$, uncertainty $\hat{\boldsymbol{\sigma}}_h$, and an action chunk $\hat{\boldsymbol{a}}_{0:H_a-1}$. Future labels after robot contact are masked unless they are generated under the same frozen applied-action sequence; this prevents an unconditional model from being scored against a world altered by its own action.

a) Clock and label construction: The logger distinguishes sensor exposure, frame availability, policy start/end, command issue, and actuator application; none is reconstructed from a nominal frame index. At training time, the

TABLE I

MECHANISM-LEVEL POSITIONING. "CONSUMED" REQUIRES AN ABLATABLE INFERENCE PATH FROM PREDICTED FUTURE TO ACTION, NOT MERELY SHARED SUPERVISION; METRIC 3-D REQUIRES INDEPENDENTLY SCORABLE PHYSICAL UNITS.

Method	History	Metric 3-D	Target SE(3)	Future consumed	Uncertainty	Feed-forward	Principal boundary
DBC-TFP [4]	✓	–	–	✓	–	✓	Future RGB; controlled simulation
PUMA [3]	✓	–	–	–	–	✓	Future-feature auxiliary loss
DynamicVLA [1]	✓	–	–	–	–	✓	Implicit history-to-action streaming
AHEAD [5]	✓	–	–	✓	✓	✓	2-D latent kinematic extrapolation
PhysMani [6]	✓	✓	–	✓	–	–	Per-scene reconstruction/optimization
LaMP [7]	✓	✓	–	✓	–	✓	Latent scene-flow prior
SE(3)-WAM (candidate)	✓	✓	✓	✓	✓	✓	Exogenous rigid-object, short-horizon scope

state at $\tau_h^\star$ is interpolated from the physics trace. Translation is linear in the base frame, whereas rotation uses $\boldsymbol{R}_j \operatorname{Exp}(\alpha \operatorname{Log}(\boldsymbol{R}_j^\top \boldsymbol{R}_{j+1}))$; object symmetry is applied only by the loss, never by altering the label. Simulated flow transports visible depth samples through persistent body identity into the same base frame. A point is invalidated when its correspondence is undefined, and an entire future is invalidated once unmatched robot contact makes it action-dependent. Training samples delays from the measured deployment support and supplies the realized ℓ_t as a continuous input. Fixed-delay replay is retained as a diagnostic, but headline manipulation uses the wall clock, where waiting for history consumes task time.

B. Timestamped Metric 3-D Encoder

For each view and valid history time, a frozen DINOv3 encoder [24] produces a 14×14 patch grid. Measured depth and camera intrinsics lift patch centers into 3-D; time-varying extrinsics map them to the robot base frame. A token combines appearance, base-frame position, view ID, true time offset $\tau_k^s - \tau_t^s$, frame age, and a validity bit. Separate tokens encode robot state, gripper state, and both issued and applied action history. A small task embedding specifies the physical contract but contains no natural-language description or future state.

A factorized transformer first performs within-frame spatial fusion and then temporal attention at matched patch/view locations. Unlike repeat padding, an explicit history-valid mask preserves the information boundary at early task steps. Base-frame 3-D positional encoding makes metric depth and camera motion available without treating depth as a third RGB channel. The encoder outputs scene tokens $\boldsymbol{X}$ and H target queries $\boldsymbol{q}_{1:H}$.

a) Information boundary: The task ID selects an evaluator and target-query slot; it does not encode object pose, motion class, destination trajectory, or event branch. Although internal records retain simulator truth for replay and supervision, the standard loader exposes only timestamped RGB-D, calibration, robot/action history, and validity masks. Future state, material parameters, motion-generator variables, RNG state, and the T5 event tape are label-only. Prediction-only P0 samples carry no action target and therefore contribute exactly zero action gradient. Normalization and uncertainty calibration are fitted on training records only.

C. Target-Centric Physical Forecast

The object queries estimate the latest visible target pose $\hat{\boldsymbol{T}}_t$ and a relative Lie-algebra trajectory $\hat{\boldsymbol{\xi}}_h \in \mathfrak{se}(3)$:

$$\hat{\boldsymbol{T}}_h = \operatorname{Exp}(\hat{\boldsymbol{\xi}}_h^\wedge)\hat{\boldsymbol{T}}_t, \quad \hat{\boldsymbol{\xi}}_h = [\hat{\boldsymbol{\rho}}_h^\top, \hat{\boldsymbol{\phi}}_h^\top]^\top. \tag{2}$$

Translation is reported in centimeters and rotation by the symmetry-aware $\mathrm{SO}(3)$ geodesic. A diagonal tangent-space log variance accompanies every horizon. In parallel, a spatial decoder predicts metric patch displacement $\hat{\boldsymbol{f}}_{i,h} = \hat{\boldsymbol{x}}_i(\tau_h^\star) - \boldsymbol{x}_i(\tau_t^s)$ and log variance. Pose and flow are complementary: $\mathrm{SE}(3)$ yields an action-ready rigid target state, whereas dense flow exposes non-target movers, depth motion, and consistency violations.

For a residual $\boldsymbol{e}$ with predicted log variance $\boldsymbol{s}$, we use the heteroscedastic penalty

$$\mathcal{L}_{\mathrm{het}}(\boldsymbol{e}, \boldsymbol{s}) = \tfrac{1}{2} \sum_j \left[\exp(-s_j) e_j^2 + s_j\right]. \tag{3}$$

The pose loss applies Eq. (3) to translation and to $\operatorname{Log}(\hat{\boldsymbol{R}}_h^\top \boldsymbol{R}_h)^\vee$ after minimizing over the declared object symmetry group. The flow loss applies it only to valid, exogenous points. A rigid-consistency term aligns target-region flow with the displacement induced by $\hat{\boldsymbol{T}}_h \hat{\boldsymbol{T}}_t^{-1}$.

D. Intervenable Physical Summary and Action Expert

Passing all spatiotemporal flow tokens to the policy would obscure the interface and increase latency. We instead form one summary token per execution horizon:

$$\boldsymbol{z}_h = g_\psi\Big(\operatorname{Log}(\boldsymbol{T}_{E,t}^{-1}\hat{\boldsymbol{T}}_h), \hat{\boldsymbol{v}}_h, \operatorname*{Pool}_i w_{i,h}\hat{\boldsymbol{f}}_{i,h}, \hat{\boldsymbol{\sigma}}_h, \tau_h^\star, \rho_{\mathrm{react}}\Big), \tag{4}$$

where $\boldsymbol{T}_{E,t}$ is the end-effector pose, $w_{i,h}$ is a learned target/mover weight, and ρ_{react} is total delay divided by the latest feasible contact time. The H tokens preserve horizon identity and uncertainty but have a fixed compute cost independent of image resolution.

The action expert predicts an H_a-step E7 chunk (body-frame translation, axis-angle rotation, and gripper). Let $\boldsymbol{c}_t = (\boldsymbol{X}, \boldsymbol{z}_{1:H})$ denote its visual–physical context. With Gaussian

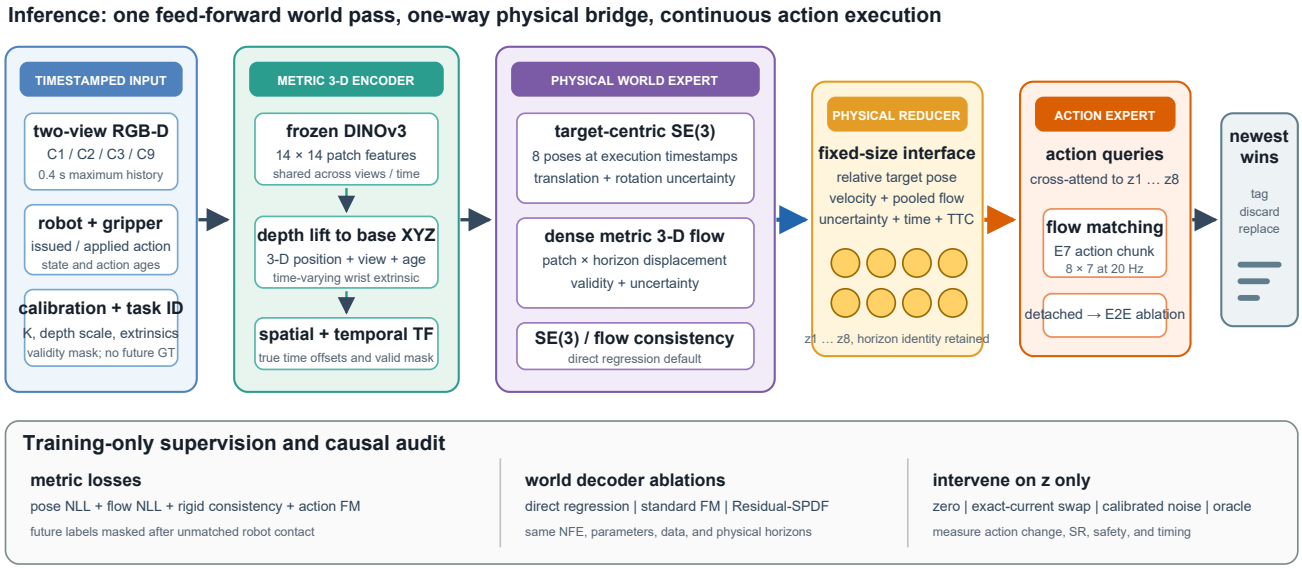


Fig. 2. SE(3)-WAM architecture. Depth-lifted frozen features are fused over true timestamps; the world expert predicts execution-aligned target SE(3), 3-D flow, and uncertainty. A target/mover reducer forms H physical tokens. The action expert retains current scene tokens but reads forecasts only through this one-way, intervenable bridge. Dashed paths are training-only.

source $\boldsymbol{a}_0$ and data chunk $\boldsymbol{a}_1$, conditional flow matching [25] uses

$$\begin{aligned} \boldsymbol{a}_\tau &= (1-\tau)\boldsymbol{a}_0 + \tau \boldsymbol{a}_1, \\ \mathcal{L}_{\text{act}} &= \|\boldsymbol{v}_\theta(\boldsymbol{a}_\tau, \tau \mid \boldsymbol{c}_t) - (\boldsymbol{a}_1 - \boldsymbol{a}_0)\|_2^2. \end{aligned} \tag{5}$$

A one-way cross-attention adapter lets action queries read $\boldsymbol{z}_{1:H}$; scene and world tokens never read action tokens. Forecast pretraining is followed by the default detached-bridge action stage, which prevents the action objective from destroying metric semantics. A one-way end-to-end stage is evaluated separately. This design gives a precise intervention point: replace $\boldsymbol{z}$ by zero, a matched-sample swap, calibrated noise, or oracle future while leaving current perception unchanged.

a) Matched information and gradient paths: The mechanism comparison is defined by three conditionals with a shared encoder, world targets, action queries, and parameter envelope:

$$\begin{aligned} \text{J-Joint:} \quad & \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}), \qquad \boldsymbol{z} \sim q_\psi(\boldsymbol{z} \mid \boldsymbol{X}), \\ \text{P-Detach:} \quad & \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}, \text{sg}[\boldsymbol{z}]), \\ \text{P-E2E:} \quad & \boldsymbol{a} \sim p_\theta(\boldsymbol{a} \mid \boldsymbol{X}, \boldsymbol{z}), \end{aligned} \tag{6}$$

where sg stops action gradients but not forecast gradients. J-Joint controls for representation shaping, P-Detach tests inference-time consumption without semantic drift, and P-E2E tests whether task gradients should reshape the physical interface. The action expert always retains $\boldsymbol{X}$, so a failure of $\boldsymbol{z}$ cannot be hidden by forcing all control through an artificial bottleneck. Conversely, interventions change only $\boldsymbol{z}$ and thus leave current perception, robot state, and action source noise fixed.

The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{act}} + \lambda_T \mathcal{L}_{\text{pose}} + \lambda_F \mathcal{L}_{\text{flow}} + \lambda_C \mathcal{L}_{\text{cons}}. \tag{7}$$

Direct world regression is the latency-oriented default. Standard world flow matching and aligned-previous-chunk Residual-SPDF are source-model ablations, not assumed contributions; either is retained only if it improves prediction and closed-loop action under equal NFE.

E. Training Protocol and Model Selection

Training is staged to keep mechanism comparisons identifiable. First, the metric encoder and world expert are trained on P0 plus valid exogenous pre-contact windows from manipulation episodes. Each loss is divided by its number of valid targets, so a long visible sequence cannot outweigh an occluded or early-history example. Minibatches are balanced over task, motion level, object family, visibility, and horizon availability rather than sampled in proportion to raw episode count. Geometric augmentation is shared by both RGB-D views and their calibration; color augmentation never changes depth or supervision. Frozen DINOv3 features may be cached, but the cache key includes image, crop, encoder, and calibration versions.

Second, the action-only current and history baselines select direct regression or flow matching on the same validation manifest. This freezes the action objective, chunk, source-noise convention, and optimizer before any explicit future is consumed. J-Joint then adds the world losses without an action edge. P-Detach initializes from that checkpoint, freezes the selected physical semantics, and trains only the reducer, bridge, and action-side parameters; P-E2E is a separately seeded continuation with all metric losses still active. No variant receives extra demonstrations, task-specific checkpoints, privileged masks, or a longer history.

World-decoder families are screened on prediction, calibration, latency, and the same closed-loop development cells.

A decoder advances only if it improves action under equal data, parameters, NFE, precision, and execution horizons; visual quality is not a selection metric. The final checkpoint rule, seeds, and Core manifest are frozen before Test-OOD, external, or real-robot evaluation. At deployment, only policy-visible inputs are loaded: simulator instance masks, target pose, future labels, and teacher feasibility scores are absent.

F. Continuous Execution

Inference runs in a latest-observation worker rather than waiting for each chunk to finish. Every output is tagged by observation and target control step. A completed chunk discards actions whose target time has passed and replaces older actions on overlapping future steps. The executor logs capture, availability, inference, issue, and apply timestamps; Eq. (1) is updated from the realized delay. This combines explicit environmental forecasting with the strongest practical lesson of streaming policies: the world never pauses for the model.

IV. DB0: Dynamic Manipulation Protocol

DB0 is a small diagnostic layer, not a claim that existing benchmarks are obsolete. It isolates observability, execution delay, metric future utility, and contact under one frozen MuJoCo 3.11/robosuite physical clock [26], [27]. Every task uses a single Franka arm, synchronized fixed and wrist RGB-D cameras at 20 Hz, 500 Hz physics, and an E7 operational-space controller. The maximum stored history is 0.4 s (nine frames); C1, C2, C3, and C9 loader views differ only in valid past context. No policy observes privileged object pose, twist, future event tape, material parameters, or natural-language hints.

Tasks. P0 predicts eight target poses at 50–400 ms without robot motion. T1-Belt intercepts a conveyor object; T1-XYZ/SO3 share centroid motion but only the latter rotates an asymmetric target, and T1-OCC adds physical occlusion. T2-TRANS/SE3 release into translating or rotating receptacles; T3-Phase/Full traverse a rotating buzz wire. T4-Sphere/Slider/Can capture rolling, sliding, or roll–slip objects leaving a ramp. After one of two striker impacts, T5-Replan must grasp the resulting keyed puck. Ordered evaluators reject touch-only, blocking, unilateral contact, freezing, and early rigid attachment.

Counterfactuals and splits. Exact-current groups share current pose, appearance, robot state, and task parameters while histories imply different future motion. Time-reversed and same-current swapped histories expose single-frame shortcuts. Test-ID supports static, constant-twist, and smoothly changing motion. Test-OOD changes exactly one predeclared object, spatial, or motion-physics factor; it is reported diagnostically and never used for model selection. Unannounced post-query events are evaluated as adaptation, not as deterministic forecast error.

Replay and outcome contract. Each episode carries a serialized base state, task factors, code/asset/controller versions, source-group identity, and a hash over the fields that must remain invariant for its paired comparison. All descendants of one object, motion seed, or source demonstration stay in one split. The accepted manifest is frozen before policy evaluation and shared by every method; policy-specific expert filtering is forbidden. The first valid observation is also the first allowed action except for T3's declared phase pre-roll, so methods cannot receive free motion history.

Success is an ordered physical event trace rather than a final pose threshold. The common prehensile contract requires opposing finger-pad contact for at least 50 ms, loss of support and 5 cm center-of-mass clearance, then a 0.5 s stable hold while the driver keeps moving. T2 analogously requires entrance-plane crossing, full collision-geometry containment, release, low relative motion, and dwell; T3 fails on calibrated wire contact. These traces localize whether a rollout failed in forecast, feasibility, approach, first contact, grasp, transport/release, or post-event adaptation, and prevent touch, pin, stop, or fixture block from being counted as success.

Metrics and statistics. Closed-loop results report task-balanced success rate (SR), hierarchical task completion (TC), completion time (CT) beside SR, safety/failure rates, and speed–delay–TTC curves. P0 and exogenous pre-contact intervals report translation and symmetry-aware rotation error, ADE/FDE, 3-D EPE, false motion on static regions, NLL, coverage, and calibration. Real-time metrics include capture-to-act age, p50/p95 inference, deadline misses, stale-prefix discards, and effective replanning rate. Headline comparisons use five independent training seeds, paired manifests, and 100 rollouts per cell; Wilson intervals and clustered paired bootstrap preserve counterfactual groups.

External validity. The project-owned matched comparison is complemented by native protocols: a single-arm DOMINO subset [3], DOM-Dyn4 with DynamicVLA [1], all normal/high PhysMani families [6], static LIBERO compatibility [28], and 180 real Franka conveyor trials. Native scores remain separate from DB0 and all compared methods are rerun under each external protocol.

V. Experiments

We evaluate five questions: **Q1**, does valid timestamped history and metric 3-D input improve dynamic action? **Q2**, does an explicit predicted future add value beyond auxiliary supervision? **Q3**, is the future physically accurate and causally consumed? **Q4**, where does the method fail across speed, delay, occlusion, contact, and compute? **Q5**, does the result transfer to native external suites, static tasks, and a real conveyor?

A. Setup and Matched Baselines

All learned DB0 methods use the same demonstrations, frozen DINOv3 features, history mask, action space, optimizer schedule, parameter envelope, controller, and paired evaluation manifest. We compare: (1) current-only Action; (2) a history-aware reactive transformer; (3) a validation-selected constant-position, velocity, or acceleration retargeter; (4) *J-Joint*, whose predicted future is auxiliary only; (5) *P-Detach*, the proposed one-way interface; (6) *P-E2E*, which

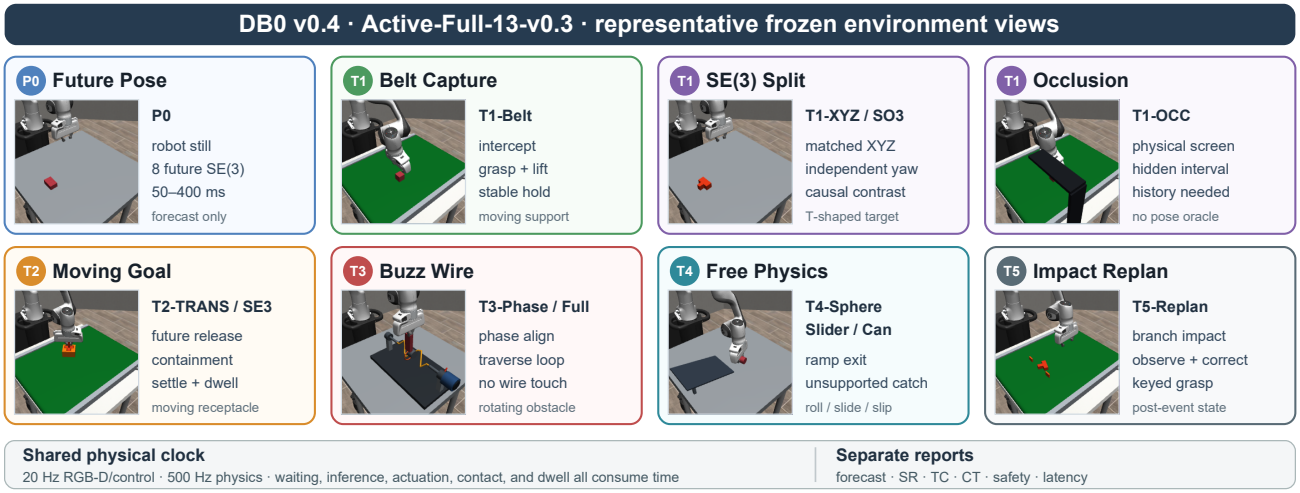


Fig. 3. DB0 v0.4 Active-Full-13-v0.3. Frozen environment-review snapshots cover P0 and all 12 manipulation cells (paired labels group variants); they are not learned-policy results. P0 is prediction-only, T1–T3 form the Core, and T4–T5 stress free physics and post-impact replanning under one physical clock.

TABLE II

PREREGISTERED CLAIM AUDIT. A CLAIM IS ACCEPTED ONLY WHEN ITS MATCHED CONTRAST AND INTERVENTION SHOW THE REQUIRED SIGNATURE WITHOUT A SAFETY OR TIMING REGRESSION. FORECAST ACCURACY ALONE CANNOT SATISFY AN ACTION CLAIM.

Claim	Matched contrast	Required signature	Reject or narrow when
History identifies motion	C1 current-only vs. C3/C9 history; exact-current and time-reversed controls	Direction/phase prediction changes with valid history and improves dynamic SR under the same action objective	History has no gain, repeat-padding leaks state, or a fixed-time/current-frame shortcut matches it
Explicit future helps action	J-Joint vs. P-Detach/P-E2E plus future-XYZ/SE(3) oracles	Consumed future improves Core SR/TC, closes oracle gap, and preserves unsafe-contact rate	Only auxiliary J-Joint improves, the oracle has no headroom, or gains vanish under matched latency
Forecast is causally consumed	Summary swap, zero, calibrated noise, and oracle within exact-current groups	Action and SR change in the preregistered direction while current observation and source noise remain fixed	Action is insensitive, changes in the wrong direction, or reacts only to scale/distribution artifacts
Orientation adds utility	T1-XYZ vs. T1-SO3; future XYZ vs. SE(3) with rotation-only interventions	Predicted and oracle SE(3) improve physical grasp/lift with centroid trajectory fixed	XYZ performs equivalently, the oracle disagrees, or texture rather than collision/grasp geometry explains the effect
Operational uncertainty	Calibrated vs. shuffled/miscalibrated variance at fixed mean forecast	Coverage and selective success improve; GO/WAIT behavior changes most near low-TTC failure boundaries	NLL improves without calibration or action/safety value
Execution-time alignment	Fixed-delay diagnostic and real wall-clock streaming at matched throughput	Benefit persists across delay/TTC and after capture-to-act age and stale-prefix handling are reported	Advantage is explained by extra compute, lower action age, missed deadlines, or different action refresh rules

TABLE III

PREREGISTERED DB0 CLOSED-LOOP SCHEMA: ONE FROZEN TEST-ID MANIFEST, FIVE SEEDS, PAIRED 95% SR INTERVALS, AND SEPARATE TC/CT. RED PLACEHOLDERS MARK THE MISSING SCIENTIFIC RESULTS.

Method	Predicted future	Action reads it?	Core SR (%) ↑	TC ↑	CT (s) ↓	T1-SO3 SR ↑	T1-OCC SR ↑	Unsafe (%) ↓
Current-only Action	None	–	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
History-reactive	None	–	[XX.X]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Analytic retarget (validation-selected)	XYZ	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
J-Joint	SE(3) auxiliary	–	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
SE(3)-WAM (P-Detach)	SE(3) **+ 3-D flow**	✓	[XX.X]	[X.XXX]	[X.XX]	[TBD]	[TBD]	[X.X]
P-E2E	SE(3) + 3-D flow	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Future-XYZ oracle	GT XYZ	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Future-SE(3) oracle	GT SE(3)	✓	[TBD ± CI]	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

permits action gradients into it; and (7) future-XYZ and future-SE(3) oracles. Direct regression and action flow matching are compared before freezing the action objective. External PUMA, DynamicVLA, and PhysMani runs use native code and scoring rather than being presented as architecture-matched ablations.

B. Estimands and Decision Rules

Table II binds each mechanism claim to a contrast fixed before viewing headline results. The primary estimand is the paired Core-SR difference between P-Detach and the strongest matched history-aware policy. Secondary outcomes are TC, CT conditional on success, unsafe contact, and per-

TABLE IV

FORECAST AND CAUSAL-INTERFACE AUDIT. FORECAST METRICS USE P0 AND EXOGENOUS PRE-CONTACT WINDOWS. INTERVENTION DELTAS ARE PAIRED WITHIN EXACT-CURRENT GROUPS; A USEFUL INTERFACE REQUIRES ACCURATE FORECASTS AND DIRECTIONALLY CORRECT BEHAVIORAL SENSITIVITY.

Model / intervention	Trans. ADE (cm) ↓	Rot. ADE (°) ↓	Flow EPE (cm) ↓	ECE ↓	Core SR (%) ↑	$\Delta\|a\|$ ↑	ΔSR (pp)	Interpretation
C1 current-only world	[TBD]	[TBD]	[TBD]	[TBD]	–	–	–	observability floor
C9 direct world	[X.XX]	[X.XX]	[X.XX]	[X.XX]	–	–	–	forecast quality
J-Joint action	same world/data envelope				[TBD]	0 (by design)	0 (by design)	auxiliary supervision
SE(3)-WAM P-Detach	predicted physical summary				[XX.X]	–	–	explicit consumption
swap summary	–	–	–	–	–	[X.XXX]	[-X.X]	exact-current causal test
zero summary	–	–	–	–	–	[TBD]	[-X.X]	information removal
calibrated noise	–	–	–	–	–	[TBD]	[-X.X]	sensitivity/calibration
oracle summary	0	0	0	0	[TBD]	[TBD]	[TBD]	controllability ceiling

TABLE V

PREREGISTERED CONTRACT, FROZEN BEFORE FINAL-RESULT ACCESS.

Item	Contract
Observation	2-view RGB-D, 224^2; robot/gripper; issued/applied actions; timestamps and validity
Clock	20 Hz sensing/control; 500 Hz physics; measured capture-to-act delay
Context	C1/C2/C3/C9 views over at most 0.4 s; no repeat padding
World target	$H = 8$ at 50–400 ms; target $\mathrm{SE}(3)$, patch 3-D displacement, uncertainty
Action	E7, $H_a = 8$; newest-wins continuous execution
Encoder	Frozen DINOv3; depth lift to base-frame XYZ; 4-layer, $d = 256$ MIN tier
Optimization	AdamW, 10^{-4}, cosine decay, 500-step warm-up, effective batch 64
Evidence	3 seeds for screening; 5 for headline; 100 paired rollouts/headline cell

task SR; they are never collapsed into a post-hoc scalar. For an explicit-future method P, history baseline B, and future oracle O, the oracle-gap fraction is

$$G_{\text{oracle}} = \frac{\mathrm{SR}(P) - \mathrm{SR}(B)}{\mathrm{SR}(O) - \mathrm{SR}(B)}, \tag{8}$$

reported only when the denominator is positive and its uncertainty is propagated by the paired bootstrap. A gain is insufficient if its interval includes a preregistered practically irrelevant region or if unsafe contact rises.

Each headline cell reports five seed-level values and 100 paired rollouts per seed; Wilson intervals describe individual SRs, while a clustered paired bootstrap resamples exact-current groups within seed for method differences. Horizon, task, visibility, and motion-factor analyses are declared strata, not opportunities to replace the Core result. Test-OOD changes one factor and is opened only after model selection. Static, external, and real-robot tracks are separate estimands. Seeds, manifests, checkpoints, evaluation code, precision, and realized timing are frozen before unblinding these tracks.

C. Closed-Loop Manipulation

Table III reports the primary matched comparison. SE(3)-WAM achieves [XX.X]% Core SR, [X.XXX] TC, and [X.XX] s CT, compared with [XX.X]% SR for the strongest matched history-aware policy. The paired SR difference is [+X.X] points (95% CI [X.X, X.X]). Unsafe-contact rate is [X.X]%. We report each task and factor rather than allowing easy cells to conceal a catastrophic regression. The explicit-future claim is retained only if P-Detach closes a preregistered fraction of the future-$\mathrm{SE}(3)$ oracle gap and does not trade success for unsafe contact.

TABLE VI

PREREGISTERED EXTERNAL, STATIC, REAL-ROBOT, AND REAL-TIME SCHEMA. BUDGETS ARE OURS; NATIVE SCORES REMAIN SEPARATE FROM DB0.

Protocol / metric	Planned N	Reference	SE(3)-WAM	Δ [95% CI]
DOMINO-SA10 SR (%) [3]	1,000	[TBD]	[TBD]	[TBD]
DOM-Dyn4 SR (%) [1]	800	[TBD]	[TBD]	[TBD]
PhysMani-Dyn16 SR (%) [6]	640	[TBD]	[TBD]	[TBD]
LIBERO-Static40 SR (%) [28]	2,000	[TBD]	[TBD]	[TBD]
Real Franka conveyor SR (%)	180	[TBD]	[TBD]	[TBD]
Inference p50/p95 (ms)	–	–	[XX.X]/[XX.X]	–
Action age p95 / miss rate	–	–	[XX.X] ms / [X.X]%	–

D. Forecast Quality and Causal Use

The selected world expert obtains [X.XX] cm translation ADE, [X.XX] degree rotation ADE, [X.XX] cm 3-D EPE, and [X.XX] calibration error (Table IV). P-Detach closes [XX]% of the oracle action gap. Within exact-current groups, swapping the physical summary changes the normalized action chunk by [X.XXX] and SR by [-X.X] points; zeroing and calibrated noise change SR by [-X.X] and [-X.X] points. These interventions are interpretable only together: high forecast accuracy without action sensitivity supports auxiliary representation learning, whereas action sensitivity to an incorrect or uncalibrated forecast is a failure mode rather than evidence of useful foresight.

We additionally stratify errors by horizon, static/moving region, visibility, pre-contact/contact state, speed, acceleration, and rotation. The T1-XYZ/SO3 pair supports an orientation claim only if predicted and oracle future-$\mathrm{SE}(3)$ both improve physical grasp/lift success over matched future-XYZ, and rotation-only interventions degrade it while centroid predictions remain fixed.

E. Latency, External Validity, and Failure Modes

At batch one, the full loop takes [XX.X]/[XX.X] ms p50/p95, with [XX.X] ms p95 capture-to-act age and [X.X]% missed deadlines. We report sensor, world, action, serialization, and controller time separately; Residual-SPDF is compared at equal NFE and is never credited with compute savings. Table VI contains native DOMINO-SA10,

DOM-Dyn4, PhysMani-Dyn16, LIBERO-Static40, and real-conveyor results.

Failure analysis uses event logs rather than qualitative anecdotes. Categories are prediction (wrong direction/phase/depth), feasibility (too-late intercept), planning/IK, first contact, unilateral grasp, slip/drop, unsafe fixture contact, stale action, deadline miss, and post-event adaptation. We retain the simplest history-aware policy if explicit prediction is statistically equivalent, and we remove any world output that neither improves action nor remains necessary for a declared downstream consumer.

VI. Limitations and Conclusion

SE(3)-WAM targets single-arm manipulation of task-specified, exogenously moving rigid objects over short horizons. It does not model language-conditioned target selection, long-horizon task planning, multi-object identity, deformables, or counterfactual contact outcomes under unseen robot actions. Dense flow after contact is scored only under matched applied actions, and confidence estimates do not by themselves constitute a safety guarantee. The target-centric interface also depends on sufficient depth quality and visible history; abrupt unobserved events require re-observation and replanning rather than impossible prediction.

Within this scope, SE(3)-WAM frames dynamic manipulation around when an action takes effect, combining timestamped metric 3-D history, explicit future $\mathrm{SE}(3)$/flow/uncertainty, an intervenable one-way interface, and continuous execution. DB0 asks whether physically correct foresight improves closed-loop behavior beyond strong history-aware streaming policies under equal information and budget. Paired results and interventions in Tables III– VI provide the answer, not forecast visualization alone.

References

[1] H. Xie, B. Wen, J. Zheng, Z. Chen, F. Hong, H. Diao, and Z. Liu, "DynamicVLA: A vision-language-action model for dynamic object manipulation," *arXiv preprint arXiv:2601.22153*, 2026.

[2] J. Tang, Y. Sun, Y. Zhao, S. Yang, Y. Lin, Z. Zhang, J. Hou, Y. Lu, Z. Liu, and S. Han, "VLASH: Real-time VLAs via future-state-aware asynchronous inference," *arXiv preprint arXiv:2512.01031*, 2025.

[3] H. Fang, S. Li, S. Wang, X. Xi, D. Liang, and X. Bai, "Towards generalizable robotic manipulation in dynamic environments," in *European Conference on Computer Vision*, 2026.

[4] Y. Zhang, R. Wang, and X. Chen, "Dynamic behavior cloning with temporal feature prediction: Enhancing robotic arm manipulation in moving object tasks," *IEEE Robotics and Automation Letters*, vol. 10, no. 6, pp. 5209–5216, 2025.

[5] S. N. Syed, A. Jakobsson, H. Hao, and J. Ichnowski, "Intercepting the future: Latent-space predictive world model for dynamic VLA manipulation," *arXiv preprint arXiv:2606.02486*, 2026.

[6] P. Yun, S. Huang, H. Li, J. Li, J. Wang, and B. Yang, "PhysMani: Physics-principled 3D world model for dynamic object manipulation," in *European Conference on Computer Vision*, 2026.

[7] X. Wang, C. Wang, Y. Xu, M. Ye, F. Zhang, J. Tian, X. Zhan, L. Zhu, C. Lu, and L. Yang, "LaMP: Learning vision-language-action policy with 3D scene flow as latent motion prior," in *European Conference on Computer Vision*, 2026.

[8] I. Akinola, J. Xu, S. Song, and P. K. Allen, "Dynamic grasping with reachability and motion awareness," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2021, pp. 9422–9429.

[9] Z. Li, J. Yang, Y. Zhao, L. Ren, X. Wu, Z. Xu, and H. Zhao, "Sim-to-real dynamic object manipulation on conveyor systems via optimization path shaping," *arXiv preprint arXiv:2508.14042*, 2025.

[10] J. Cai, L. Ling, S. Chu, Z. Liu, J. Kang, Z. Liang, W. Xu, Y. Mao, W. Zhang, X. Yang, R. Ying, R. Zheng, and Y. Mu, "AHA-WAM: Asynchronous horizon-adaptive world-action modeling with observation-guided context routing," *arXiv preprint arXiv:2606.09811*, 2026.

[11] Z. Zhang, Y. Jiang, Q. Cheng, X. Li, S. Liu, D. Zhao, P. Luo, W. Zhou, and H. Yu, "GaussianDream: A feed-forward 3D gaussian world model for robotic manipulation," *arXiv preprint arXiv:2605.20752*, 2026.

[12] B. Li, X. Yin, M. Lin, Y. Zhang, and D. Xu, "EgoWAM: World action models beyond pixels with in-the-wild egocentric human data," *arXiv preprint arXiv:2607.08436*, 2026.

[13] T. Yuan, Z. Dong, Y. Liu, and H. Zhao, "Fast-WAM: Do world action models need test-time future imagination?" *arXiv preprint arXiv:2603.16666*, 2026.

[14] J. Chen, K. Wang, K. Chen, S. Chen, F. Gao, W. Tang, Z. Li, W. Liu, Z. Yao, B. Li, Y. Xu, and C. Yu, "LaWAM: Latent world action models for efficient dynamics-aware robot policies," *arXiv preprint arXiv:2606.15768*, 2026.

[15] H. Jiang, Y. Zou, B. Liang, B. Liu, F. Meng, and S. Liu, "FuturERTC: Real-time robot execution with anticipatory-conditioned action chunking," *arXiv preprint arXiv:2607.24008*, 2026.

[16] L. Qiu, Y. Li, Y. Chen, Y. Ge, Y. Ge, and X. Liu, "Making foresight actionable: Repurposing representation alignment in world action models," *arXiv preprint arXiv:2606.12217*, 2026.

[17] Y. Liu, P. Sun, S. Li, Y. Xie, L. Zhang, X. Chao, S. Dong, F. Chen, X.-P. Zhang, and W. Ding, "OA-WAM: Object-addressable world action model for robust robot manipulation," *arXiv preprint arXiv:2605.06481*, 2026.

[18] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," *arXiv preprint arXiv:2304.13705*, 2023.

[19] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. C. M. Burchfiel, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," in *Proceedings of Robotics: Science and Systems*, 2023.

[20] Y. Ze, G. Zhang, K. Zhang, C. Hu, M. Wang, and H. Xu, "3D diffusion policy: Generalizable visuomotor policy learning via simple 3D representations," in *Proceedings of Robotics: Science and Systems*, 2024.

[21] W. Huang, Y.-W. Chao, A. Mousavian, M.-Y. Liu, D. Fox, K. Mo, and L. Fei-Fei, "PointWorld: Scaling 3D world models for in-the-wild robotic manipulation," *arXiv preprint arXiv:2601.03782*, 2026.

[22] J. U. Kreber, L. Mack, and J. Stueckler, "Learning action-conditional and object-centric gaussian splatting world models for rigid objects," *arXiv preprint arXiv:2606.01950*, 2026.

[23] H. Zhi, P. Chen, S. Zhou, Y. Dong, Q. Wu, L. Han, and M. Tan, "3DFlowAction: Learning cross-embodiment manipulation from 3D flow world model," *arXiv preprint arXiv:2506.06199*, 2025.

[24] O. Siméoni, H. V. Vo, M. Seitzer, F. Baldassarre, M. Oquab, C. Jose, V. Khalidov, M. Szafraniec, S. Yi, M. Ramamonjisoa, *et al.*, "DINOv3," *arXiv preprint arXiv:2508.10104*, 2025.

[25] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, "Flow matching for generative modeling," in *International Conference on Learning Representations*, 2023.

[26] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 5026–5033.

[27] Y. Zhu, J. Wong, A. Mandlekar, R. Martín-Martín, A. Joshi, K. Lin, A. Maddukuri, S. Nasiriany, and Y. Zhu, "robosuite: A modular simulation framework and benchmark for robot learning," *arXiv preprint arXiv:2009.12293*, 2020.

[28] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, "LIBERO: Benchmarking knowledge transfer for lifelong robot learning," *arXiv preprint arXiv:2306.03310*, 2023.